

Briefing Document: Fork-Sync-All Repository Management System

Executive Summary

The **fork-sync-all** project serves as the centralized "control plane" for the automated management of git-based organizations and repositories, primarily within the Interested-Deving-1896 GitHub organization. This system orchestrates a complex network of automation—including approximately 180 GitHub Actions workflows—designed to keep multiple GitHub organizations and GitLab groups in synchronized operation. The system addresses the operational anxieties associated with large-scale repository maintenance by automating critical tasks such as fork synchronization, README generation, badge injection, upstream tracking, mirroring, and release management. Key capabilities include a multi-tier mirror chain, an opt-in Over-the-Air (OTA) update system, and integrated AI agent cost tracking. Operating without compiled artifacts or Node.js runtime dependencies, the system relies on a robust foundation of Bash scripts and environment variables to ensure consistency across distributed version control systems.

Core Operational Mechanics

Functional Infrastructure

The project is built entirely on Bash scripts and GitHub Actions. It is notable for its lack of compiled artifacts, build steps, or Node.js dependencies, ensuring that every workflow executes directly against the shell scripts located in the `scripts/` directory.

- **Communication:** Scripts interact via environment variables and exit codes.
- **Shared Helpers:** Utility scripts located in `scripts/includes/` provide standardized GitHub API access (`gh-api.sh`), quota budgeting (`budget.sh`), and run instrumentation (`quota-instrument.sh`).
- **Mirror Chain:** The system maintains an outward mirroring flow:
Interested-Deving-1896 → OpenOS-Project-OSP →
OpenOS-Project-Ecosystem-OOC → GitLab.

System Scale

The project manages a massive footprint across several platforms:

Metric	Current Scale
Total Workflows	~180
Registered Upstream Imports	157
Template Consumers	82 Repositories
GitLab Subgroups	14
GitLab Repositories Mirrored	225

Workflow Categorization and Automation Groups

The automation suite is divided into approximately 20 functional groups to manage specific aspects of the repository lifecycle.

Group	Key Functions
Mirror Chain	Orchestrates outward mirroring across GitHub orgs and GitLab.
Fork & Import Sync	Synchronizes upstream forks and manages registered imports from Bitbucket, Gitea, and Sourcehut.
PR Governance & Trust	Manages PR gates, labeling, auto-merging (vouch), and rebasing.
README Management	Automates the creation, translation, and validation of README files and badges.
CI & Failure Resolution	Includes rate-limit reruns, failure resolvers, and runner status monitoring.
OTA System	Delivers updates to opted-in forks

through a release delivery and reconciliation architecture. || **Quota & Queue Management** | Monitors REST call costs, deduplicates queued runs, and manages "quota floors." || **Accessibility** | Conducts WCAG audits, screen-reader scans, and generates Braille and audio outputs. || **AI & Cost Tracking** | Logs session costs and synchronizes pricing for AI agent tasks using OCU budgeting. |

Configuration and Governance

Primary Configuration Files

The system's behavior is governed by machine-readable YAML and JSON configurations:

- **config/template-manifest.yml** : Defines profiles for template synchronization (full, mirror, infra-core, or standalone).
- **config/workflow-priority-tiers.yml** : Establishes cancellation priorities, where Tier 1 workflows are protected from cancellation and Tier 4 are canceled first.
- **config/workflow-quota-costs.yml** : Provides REST call estimates per workflow to drive quota pre-flight checks.
- **config/agent-cost-profiles.yml** : Contains eight variants and ten complexity tiers for AI agent cost tracking.
- **registered-imports.json** : Tracks 156-157 upstream repositories kept in ongoing synchronization.

Quota and Rate Limit Management

The system is highly sensitive to GitHub's REST API limits. Both SYNC_TOKEN and GH_SYNC_TOKEN share a single pool of **5,000 requests per hour** .

- **GitHub REST/GraphQL**: 5,000 requests/points per hour, resetting at the top of the hour.
- **GitLab REST**: 2,000 requests per minute.
- **Efficiency**: Fetches from raw.githubusercontent.com do not count against the API quota.

Ecosystem Integration: GitLab Subgroups

The system mirrors 225 repositories into 14 specialized subgroups under gitlab.com/openos-project. Key subgroups include:

- **agnostic-api_deving (29 repos)**: Focuses on virtual filesystems, AI/LLM adapters, and OS-compatibility layers.
- **incus_deving (49 repos)**: Dedicated to Incus container and VM tooling.
- **yaml-tooling_deving (34 repos)**: Manages linters, schema validators, and GitHub Actions tooling.
- **accessibility_deving (9 repos)**: Focuses on screen readers, Braille output, and WCAG auditing.

Accessibility and Security Compliance

The project integrates high-level auditing and security protocols:

- **Automated Auditing**: The check-accessibility.yml workflow utilizes espeak-ng for audio overviews and liblouis for Braille output, ensuring compliance with WCAG 2.1 AA standards.

- **Security & Compliance:** Automation includes Software Bill of Materials (SBOM) generation, CodeQL analysis, vendor audits, and token rotation.
- **Identity Reconcile:** An agnostic identity asset system ensures consistent branding and assets across all mirrored platforms.

Key Operational Quotes

"When Git Platforms Give You Anxiety Attacks, Who Are You Going To Call?

Fork-Sync-All!" "All automation is implemented as Bash scripts in scripts/ and GitHub Actions workflows in .github/workflows/. There are no compiled artifacts, no Node.js runtime dependencies, and no build step — every workflow runs directly against the shell scripts."